

Enhancing Wildlife Tourism: A Comprehensive System for Safari Experience Optimization

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Declaration

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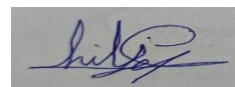
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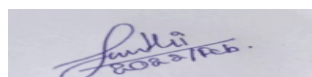
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Abstract

Although wildlife tourism presents a unique opportunity to be in touch with nature, it also raises questions regarding both animal and human welfare and safety. The goal of this project is to create a smart system that tracks the whereabouts of animals in real time, prioritizes safety, and makes use of cutting-edge technology to improve the safari experience. Voice help for passengers, on-the-spot animal identification, and risk assessment based on historical events linked to certain species are all features of the system. It will also track the whereabouts of animals through the image, enabling safari drivers to arrange safer and more effective itineraries and, in the end, giving guests a more fun and safe experience.

This project is unique in that it combines real-time location data, machine learning, and voice assistance to create an intelligent and interactive safari. The ultimate result will be a fully functioning mobile application that aims to enhance wildlife tourism by raising safari tourists' satisfaction, fulfillment and security.

Keywords: location tracking, safety monitoring, machine learning, voice help, wildlife tourism, and animal identification.

Acknowledgement

I express sincere gratitude to my dissertation supervisors, Dr. Dharshana Kasthurirathna and Ms. Samanthi Siriwardana., for their unwavering support. Thanks to the Faculty of Computing and Department staff and fellow students for their contributions, provided valuable support. I also thank my family and friends for their unwavering encouragement during this research journey.

Table of Contents

Declaration.....	3
Abstract.....	4
Acknowledgement	5
List of Tables.....	8
List of Figures	9
1.0 INTRODUCTION	10
1.1 Background & Literature Review	10
1.1.1 Background.....	10
1.1.2 Literature Review.....	12
1.2 Research Gap	14
1.3 Research Problem	16
1.4 Research Objectives.....	19
General Objective	20
Specific Objectives	20
2.0 METHODOLOGY	24
2.1 Methodology	24
2.1.1 Method Used (ML Model, Data Flow, etc.).....	27
2.1.2 Data Collection and Preprocessing	27
2.1.3 Animal Identification Model.....	29
2.1.4 Harmfulness Evaluation Model	31
2.1.5 Real-Time Animal Tracking and Location Estimation.....	33
2.1.6 Voice Assistant Integration.....	36
2.2 Commercialization Aspects of the Product.....	36
2.2.1 Market Potential and Target Users	37

2.2.2 Product Positioning and Value Proposition.....	37
2.2.3 Monetization and Revenue Models.....	38
2.2.4 Scalability and Future Expansion	38
2.2.5 Technical Feasibility for Market Use	39
2.3 Testing & Implementation.....	39
2.3.1 Testing the Animal Identification Model	40
2.3.2 Testing the Harmfulness Prediction Module	41
2.3.3 Location Estimation and Sensor Accuracy	42
2.3.4 Voice Assistant Response Evaluation.....	43
2.3.5 System Integration and End-to-End Testing	43
2.3.6 Usability Test and Feedback	44
2.3.7 Final Deployment.....	44
3.0 RESULTS & DISCUSSION	45
3.1 Results.....	45
3.2 Research Findings	49
3.3 Discussion	52
4.0 CONCLUSION.....	55
5.0. REFERENCE.....	58
6.0 APPENDIX	60

List of Tables

Table 1-1 Significance Table..... 14

List of Figures

Figure 1-1: System Flow Chart.....	19
Figure 1-2: System diagram for the component.....	23
Figure 1-3: System diagram for the whole system	24
Figure 2-1: Animal Identification Model visualization.....	60
Figure 2-2: Aggressive detection Model visualization.....	6032
Figure 3-1: Aggressive Animal Popup	60
Figure 3-2: Result Screen UI	6046
Figure 3-3: User Map UI.....	6047
Figure 3-4: Object Detection	6048
Figure 6-2: Existing Systems	60
Figure 6-2: Questioner1	60
Figure 6-3: Questioner2	61
Figure 6-4: Turnitin Report.....	602

1.0 INTRODUCTION

1.1 Background & Literature Review

1.1.1 Background

Wildlife tourism is a unique overlap of adventure, learning, and conservation that offers individuals an experiential way of relating to biodiversity and natural landscapes. Among the most captivating forms of wildlife tourism is safari travel, which offers visitors the chance to see free-ranging animals in the wild, often in national parks and protected reserves. Safari tourism in countries like Sri Lanka, Kenya, Tanzania, and South Africa has been a big source of regional and international income from tourism that generates hundreds of thousands of annual tourists. It not only introduces national income by way of tourism but also brings about greater recognition of the environment and more commitment to wildlife protection.

But with the new fashion for safari-type travel comes a complex set of issues. Most pressing is the safety of tourists, particularly when near possibly dangerous animals. A vast majority of safari travelers lack the experience or training necessary to effectively identify animals and understand animal behavior. Ignorance like this can lead to lost educational opportunities, as well as fatal misassumptions. For instance, misreading the body postures of an elephant or overlooking a territorial leopard's menace can cause severe or even fatal accidents. Such incidents have been witnessed in several parks, including Sri Lanka's Yala National Park, where human-animal conflicts often take place owing to faulty situational awareness.

Second. The traditional safari tourism is highly dependent on guide and driver skill, who will read animal behavior, maintain good distance, and explain things to tourists. While most guides are competent, this human-based model limits scalability and consistency. Tourists who go on do-it-yourself tours or on tours with relatively inexperienced guides will not receive the amount of information or safety guidance required for a fun and safe experience. Inadequate standardized digital tools to provide real-time guidance only exacerbates this.

Such gaps can be filled and redefine tourist interactions with wildlife by technological interventions. Mobile computing and smartphones have provided an opportunity to offer real-time, smart assistance in remote locations. Machine learning (ML), and particularly deep learning, has proven very promising in the fields of image classification and object detection, and therefore it is possible to correctly identify animals through mobile cameras. At the same time, GPS and magnetometer sensor integration offers spatial awareness, with the capacity to estimate positions of animals from the user position and orientation. Voice assistant technology can be an extension of this system by providing instructions based on voice command, which is critical when users are required to maintain visually focused engagement with the environment.

This project proposes the development of a mobile solution called "Enhanced Wildlife Identification" with the vision to transform the safari experience through intelligent automation and real-time interaction. The proposed application is intended to be useful to both tourists and jeep drivers through the following essential features:

- Real-time animal identification via mobile camera input
- Predictions of harm from history-based incident and behavior data
- GPS-based animal location estimation and device orientation
- Voice-command, context-sensitive free-hand guiding and alerting
- Easy to use, intuitive user interface to view results, receive instructions, and track sightings

By incorporating all these attributes on a single platform, the system addresses the most critical loopholes in present wildlife tourism software. By integrating all these functionalities within a single platform, the system addresses the vital gaps in currently available wildlife tourism software. This app, as opposed to most identification apps developed on user-uploaded principles, is specifically tuned for performance for real-time field use. While it allows intelligent animal identification, it also functions as an

safety manual, allowing the user to take quick, fact-based decisions when interacting with wildlife.

By doing so, the Enhanced Wildlife Identification system is set to transform safari from passive looking to active discovery, empowering individuals with the information, ability, and self-assurance to engage responsibly and safely with wildlife. This innovation is part of broader trends in smart conservation technology and smart tourism, setting a precedent for future development in eco-tourism and digital wildlife education.

1.1.2 Literature Review

In the recent past, technology in machine learning and computer vision has revolutionized wildlife research as well as tourist practice. Many research studies have highlighted animal detection, identification, and behavioral forecasting based on deep learning models, particularly Convolutional Neural Networks[3] (CNNs) and transfer learning techniques.

Amrutha et al. (2024) proposed an unsafe animal detection system using image processing[2] where they highlighted venomous snake and other dangerous wildlife detection on farms. Even though the study proves the effectiveness of visual feature-based recognition, its implementation is limited and not suitable for real-time safari applications.

However, another research carried out by Maduranga and Sithara (2023) was on wearable sensors and cellular networks' use in delivering real-time location estimation of animals, where the benefits of merging electromagnetic data and GPS to track animal mobility were presented[1]. However, there was no user interaction interface, as well as animal identification or user protection.

Species classification, Prudhivi et al. (2023) proposed a deep learning-based animal species classification model that demonstrated the efficiency of CNNs in image classification[5]. This serves to support the design of the animal identification module in our system with initial concepts in architecture choice and preprocessing of data.

Current systems like Wildbook and iNaturalist [6] have made developments in wildlife tracking and citizen science. Wildbook Lynx[8], in contrast, utilizes computer vision for animal recognition of individuals through pattern recognition, primarily utilized for conservation and research applications. It is still only applied to a small number of species and does not have real-time or voice interaction capabilities. iNaturalist, in contrast, offers user input support for entry of plant and animal observations to be identified but relies on user contribution and is not designed with safari-based real-time assistance in mind. In contrast, the proposed system aims to fill these gaps by integrating all core functionalities into a single mobile application:

- Real-time image-based identification
- Historical aggression data analysis
- Location estimation using GPS and sensor data
- Voice assistance for hands-free support

The novelty lies in its holistic integration of safety, navigation, identification, and user feedback, making it uniquely suited for dynamic, real-time wildlife tourism scenarios.

Feature	Proposed Mobile App	Wildbook (Lynx)	iNaturalist	Merlin Bird ID
Integrated Real-Time Animal Identification	Yes	No	No	No
Location Tracking Integration	Yes	No	No	No
Harmfulness Assessment of Animals	Yes	No	No	No
Automatic Data Collection (From Facial Image)	Yes	No	No	No

Real-Time Animal Identification with Historical Data	Yes	No	No	No
Voice Assistance for Navigation and Safety	Yes	No	No	No
Real-Time Updates and Alerts	Yes	No	No	No

Table 1-1 Significance Table

1.2 Research Gap

Despite increasing technology incorporation in wildlife observation and eco-tourism, there exists an enormous and largely untapped demand for the delivery of an integrated, intelligent system that is tailored to the real-time requirements of safari habitats. Even though lots of scholarly work and business solutions have been created in an endeavor to solve disconnected problems such as species identification, animal monitoring, and environmental logging, they function independently. Most lack integrated, context-sensitive decision-making capabilities and are unsuitable to the safety-critical, high-speed environment of safaris. As such, there is a pressing need for an intelligent, mobile-first platform that integrates species identification, estimation of harmfulness, positioning, and voice-based interaction into a single cohesive tool suitable for both tourists and safari operators.

Existing wildlife identification systems such as iNaturalist, Merlin Bird ID[7] provide users with the ability to upload photos and receive classification feedback through either cloud-based inference models or community validation mechanisms. Although adequate for typical educational or citizen science use cases, these systems are not appropriate for high-risk, high-velocity environments such as wildlife safaris. They typically depend on constant internet connectivity and consist of multi-step manual interfaces, which can cause risky lags in a scenario where tourists may be unknowingly approaching aggressive or territorial creatures. Additionally, the lack of real-time threat evaluation and safety protocols renders these tools ineffective in providing timely, actionable help.

Technological devices such as GPS collars, RFID [4]tags, and satellite telemetry are used primarily in professional wildlife surveys and conservation studies in wildlife monitoring. Although these technologies have been effective at providing useful ecological data, they are unsuitable for the average safari tourist. Most significantly, these solutions require bespoke infrastructure and are not suitable for mobile-based, user-driven implementation. Therefore, the gap of location awareness is still large, and travelers still do not have real-time spatial information about nearby animals, their trajectory paths, and their relative direction and distance.

The existing research identifies a clear opportunity to bridge this gap by introducing an innovative method of animal location estimation based on user-provided images. Unlike traditional wildlife tracking with tagged animals, this system leverages the captured image, device orientation, and GPS metadata to estimate the location of the animal in the field of view of the user. By extrapolating direction vectors from compass azimuth and estimating distance with algorithms for focal length and image scaling, the system can extrapolate a real-time directional location pointer. This technique offers an infrastructure-free, field-based solution that allows users to map and track visually their movements without the necessity for special apparatus or pre-marking of animals. It not only enhances spatial awareness, but also results in safer movement and better decision-making, especially under low-visibility conditions or hazardous terrain.

The other inherent shortfall in current applications is a lack of general safety assessment methodology. While certain studies, such as those conducted by Amrutha et al. (2024), are species-specific threat detection (i.e., venomous snake detection in agricultural environments such systems tend to be limited to niche use cases and do not seek to incorporate larger wildlife datasets or behavioral modeling. They do not consider contextual factors such as time of day, animal distance, or incident history in the area, all of which are required for uniform and accurate risk prediction. Lack of such intelligence denies users with appropriate tools to evaluate the threateningness of an animal seen, leaving them open to the threat of danger underestimation or neglect.

Another equally disturbing gap in current tools is the lack of intelligent voice interaction. In safari tourist use, where the users will quite frequently be on the move and need to maintain visual attention on their surroundings, voice access to safety information becomes essential. Most current applications do not offer real-time voice commentary or use pre-recorded static audio files that are not sensitive to environmental situation and user activity. Inadequate or absent voice-controlled feedback severely reduces usability, particularly in high-risk situations or when human-computer interface mechanisms may be dangerous or even impracticable.

Generally, the current technological infrastructure has a disjunctive and ineffective method of addressing wildlife tourists' safety and information requirements. There is a clear and urgent requirement for the delivery of an integrated, real-time, mobile application which not only identifies animals, but also evaluates their likely danger, calculates their distance through user-supplied photographs, and communicates this through voice commands. This paper addresses this requirement by delivering an integrated, intelligent mobile solution for users to make rapid, intelligent, and safe choices in fast-changing wildlife scenarios.

1.3 Research Problem

Wildlife tourism, particularly when it involves guided safaris, has experienced unparalleled growth over the past ten years. Such experiential vacations not only allow tourists to see animals in their natural habitats but also serve as an important vehicle for conservation education and biodiversity promotion. However, despite safari tourism becoming increasingly mainstream and ecologically relevant, the integration of intelligent, real-time technological support into such vacations remains in its early stages. Travelers and guides are frequently ill-equipped with apparatus that supports real-time identification, risk awareness, and data-oriented navigation in tough and often unstable wildlife environments.

The most elementary issue in this field is that safari users are unable to properly and timely identify the animals they encounter. This lack of ability to recognize them upon contact affects both learning gains and security. Visitors who are new to animal

behavior can misunderstand signs of aggression or territoriality and thereby pose a higher risk of stirring up hostile wildlife. For example, underestimating the mood of leopards or elephants can lead to extreme human-wildlife conflict. It is also worsened by the lack of automated platforms that can perceive visual, spatial, and contextual information in real time to provide actionable feedback.

Current technology such as field guides, static mobile apps, or crowd-sourced classification portals like iNaturalist and Wildbook Lynx are useful for record-keeping and retrospective analysis but lacking in areas of response time, automation, and warning safety. These portals need to be uploaded manually with images and entail significant user interaction, making them unsuitable for rapid field-level decision-making. In addition, they do not facilitate behavior-based estimation of danger or dynamic user context such as animal proximity or approach direction, both of which are essential in real safari environments.

Beyond identification, a major unmet need is the ability to track real-time animal location from user-shot imagery. Traditional wildlife tracking involves dependence on GPS collars or drone surveillance, technologies that are too expensive for mass tourists or deployable in extensive safari areas. This research aims to solve that issue through a new approach that estimates animal location from the obtained image, GPS, and device orientation. By integrating focal length estimation with compass azimuth and image scale, the system can estimate the location of the animal in space and present it on a dynamic map interface. Besides supporting tourist situational awareness, this capability also enables tour guides to obtain predictive insight into movement patterns of animals, increasing the chances of future sightings and facilitating safer route planning.

Such locational tracking of animals based on photographs further enables temporal logging of whereabouts of wildlife to develop a database to facilitate monitoring of movement patterns, identification of risk zones, and recommending tourists to safer observation locations. This transforms a regular tourism app into a support-for-conservation and predictive-advisory tool. The system developed performs dual reactive and proactive functions reactively by issuing warnings of impending danger,

and proactively by guiding tourists towards optimized wildlife observation experiences.

Another crucial feature missing in existing systems is voice-supported, hands-free guidance. Most mobile wildlife apps utilize visual output or static text, which is impractical while on safari where tourists must be visually focused and physically mobile. Incorporating smart, real-time voice guidance into the app enhances accessibility, reduces cognitive load, and supports multi-tasking, thereby enhancing user safety and comfort. All these voice assistance functionalities may offer instantaneous feedback when animal detection occurs, alert users of high-risk animals in proximity, and mimic having a virtual guide in tours, particularly in self-driven or unguided safaris.

Also, a lack of single systems that consolidate such capabilities within one intelligent interface is a clear roadblock to technological progress here. There is no platform today that unifies on-device animal identification, threat behavior analysis, direction-aware geolocation, and smart voice interaction into one cohesive experience designed for safari usage. This fragmentation makes end-users and guides rely on several tools, inducing friction and restricting the use of real-time application.

Given these limitations, the primary research question is articulated as follows,

"How do we design and deploy an intelligent, mobile-centric system that enables real-time animal identification, predicts the threat of identified species from historical and context-based information, estimates and traces animal locations through user-contributed imagery, and offers hands-free voice-guidance to enhance safety, vigilance, and predictive planning for wildlife safari tourists and guides?"

To address this challenge involves developing a converged solution from machine learning, geospatial data analysis, computer vision, and natural language processing. The system not only improves the safety and educational engagement of tourists but also allows guides to derive usable information, increases predictability when monitoring animals, and is helpful for long-term monitoring programs for wildlife. With an emulated simulation of the operation of an expert virtual tour guide, this

product refigures the traditional safari experience as an energetic, data-driven, and security-focused adventure

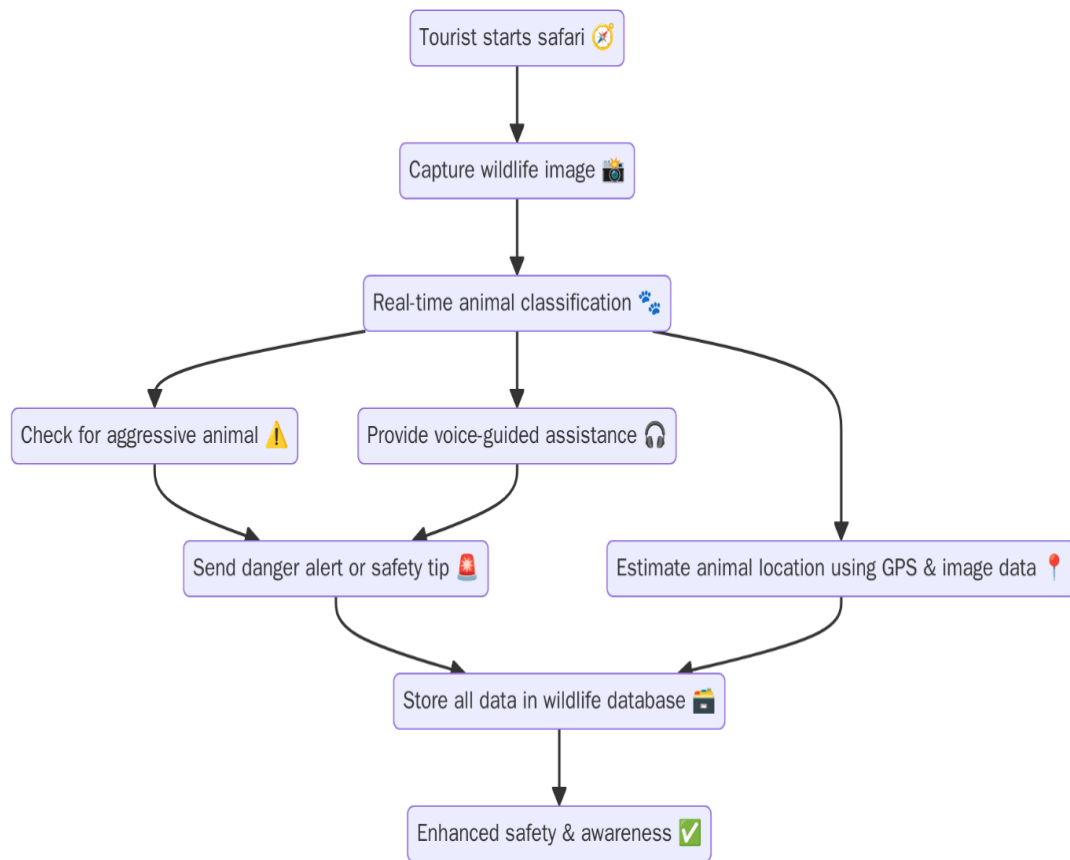


Figure 1-1: System Flow Chart

1.4 Research Objectives

This research is aimed at developing an intelligent and holistic mobile application that assists wildlife tourists, particularly safari tourists, by providing them with real-time information related to the animals they encounter. The project is founded on the pressing need for a system that not only accurately identifies animals but also predict their likely threat level, locates them in the field, and gives useful alerts and instructions through voice interaction. By integrating cutting-edge technologies like

image-based recognition, behavioral risk assessment, GPS tracking, and voice assistance, the system tries to improve the overall experience and safety of the users within the dynamic safari environment.

The grander vision for this project is to transcend the technological differences between current wildlife tourism tools by coming up with a one-stop-shop platform where the user can receive accurate, timely, and actionable information throughout the process of real-time interaction with wild animals. In order to preserve this vision, general and specific objectives have been established as follows.

General Objective

To design and deploy an integrated mobile application providing real-time animal identification, harmfulness estimation, dynamic location tracking, and voice-enabled hands-free assistance to enhance safety, situational awareness, and user experience in wildlife safari parks.

This general objective encompasses the general project theme of providing an integrated real-time solution for wildlife identification and risk mitigation through mobile technologies.

Specific Objectives

1. To develop an animal identification system in real-time using machine learning models trained on thermal and photographic images of wildlife.

This objective includes the fundamental functionality of the system—real-time species identification. A CNN-based model will be trained on high-quality and labeled images of animals in safari parks such as Yala in Sri Lanka. The model will be very accurate in classifying animals based on identifying based on certain visual characteristics such as body structure, texture. This module will be designed to execute on-device or with minimal cloud-based APIs to allow for real-time response. Model accuracy and latency will be subjected to extensive field-testing.

2. To create an image-assisted, direction-sensitive animal position estimation system for real-time positioning in safari environments.

This purpose is intended to extend beyond typical GPS tracking in offering a different method that incorporates images taken by the participants in order to estimate and track the location of the animals seen. Rather than use animal-borne equipment like out of reach and inconvenient to most safari participants GPS collars the process takes advantage of onboard sensor (camera, GPS) and smartphone computations and model results in order to estimate the location of the animal upon sighting.

When the user snaps an animal through the mobile app, the system will take the needed geospatial metadata like the current GPS location of the user and facing direction (compass direction) of the device. The system will then apply trigonometric calculation and distance estimation based on focal length, image scale, and object size in view to calculate estimated coordinates of the location of the animal.

This direction-based approximating enables the system to overlay the animal location onto a real-time map interface so that both users and guides can view current animal sightings contextually and obtain an updated, real-time experience of place. These sightings over time will get logged and predicted to generate an animal movement heat map, which provides greater probability regarding upcoming sightings and enables guides to make strategic plans way ahead or locate areas of greater activity.

With picture taking as the foundation of animal tracking rather than hardware tagging or external equipment this method offers a viable, scalable, and inexpensive alternative for the mass safari traveler. It allows users to record, revisit, and analyze the spatial wildlife experiences, enhance safety and experience without disrupting the animals' natural behavior, and promote more effective resource allocation.

3. To create a harm evaluation module that scores animals based on behavioral and past incident data.

This module corrects one of the primary safety features by identifying the threat level potential of the scored animal. Episodes of violent encounters, proximity-elicited attacks, or territorial violence will be counted into an official database. Animals will be scored into danger levels safe, cautionary, or dangerous based on this data. Machine learning classifiers such as decision trees or logistic regression models can be used to make this judgment automatically on contextual attributes such as species and similarity with aggressive dataset using a feature class file and with every image comparing its features with feature class file and have a similarity score.

4. To incorporate an in-real-time voice direction system so that one can commentate identification results, warn users of possible risk, and direct.

Due to the external nature of the safari experience, attaining this objective is vital in order to facilitate the app to become functional and effective at hand-free usage. The voice assistant will offer spoken summaries of the identified animal (e.g., species name, scientific name, behavior), offer danger alerts (e.g., "Leopard seen keep a distance"), and assist with safety guidance away from or towards animal spotting via voice-guided navigation directions. Technologies such as Google Cloud Text-to-Speech or TTS libraries will be evaluated for the best implementation under field conditions.

5. To deploy and implement a bundled mobile application combining the modules above into an easy-to-use interface optimized for running in safari.

This goal bundles all the rest of the modules into one great mobile solution brought about through technology such as React Native, MongoDB, Android SDK, and Python-based APIs. The app shall possess a user-friendly and simplified interface, off-line fallback features, as well as engaging elements such as maps, voice toggles, and image upload or camera UI. Reducing latency, keeping devices' resources intact, as

well as rendering smooth user interactions even under conditions of low band-width common within national parks, will be achieved.

Together these objectives ensure the envisioned system is not just another passive tracking device or encyclopedia of wildlife but a context-aware and safety-aware guide for wildlife tourists. Every objective adds a vital component of functionality that works toward the common aim of reducing human-wildlife conflict, enhancing awareness, and creating an interactive and informed safari experience.

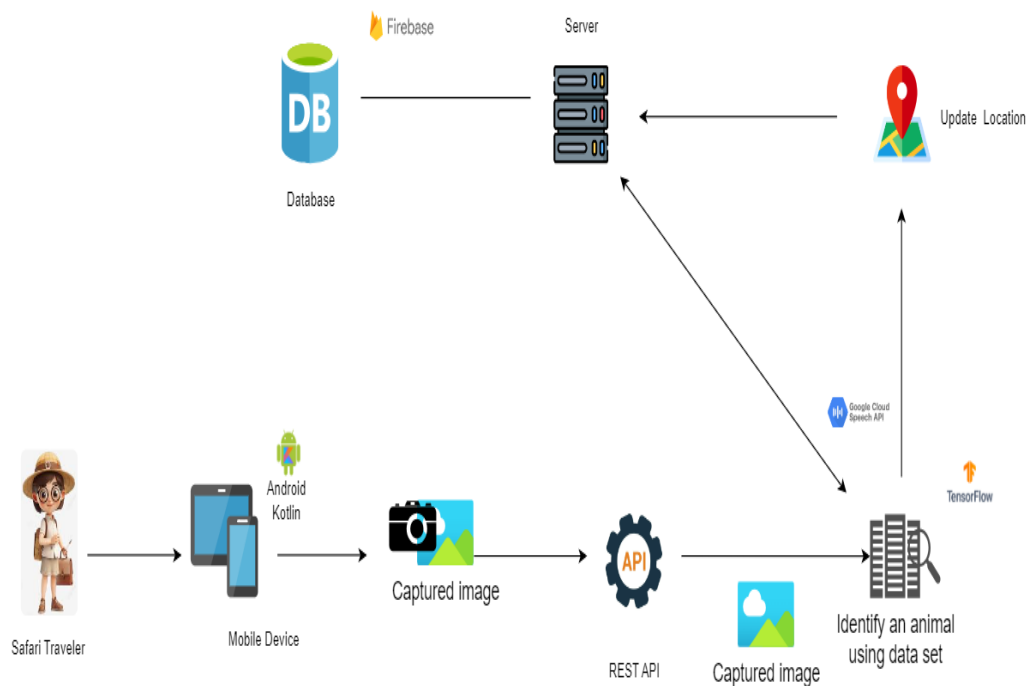


Figure 1-2: System diagram for the component

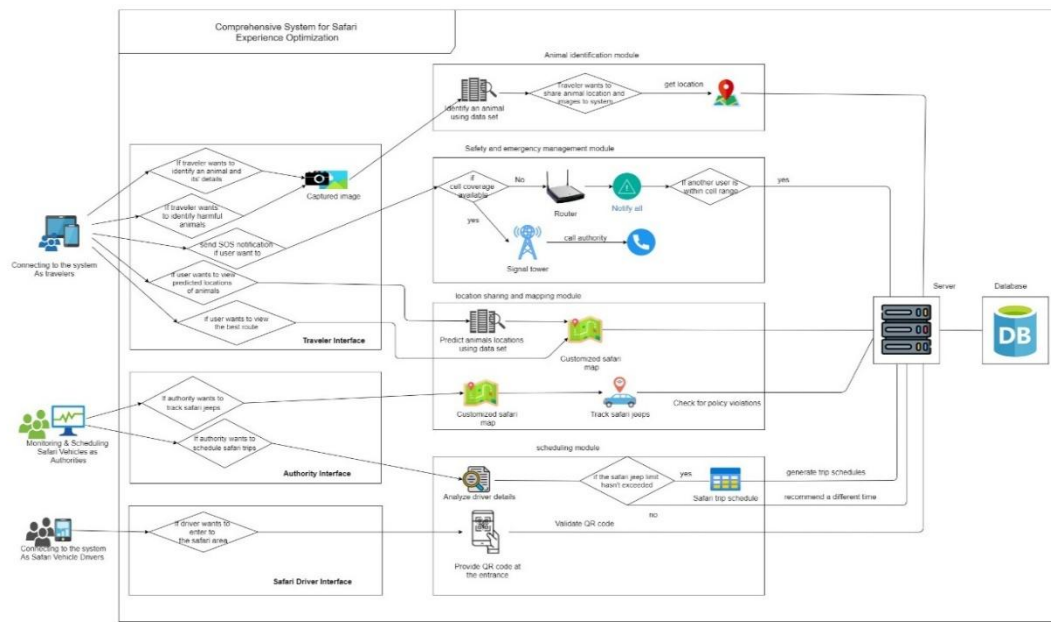


Figure 1-3: System diagram for the whole system

2.0 METHODOLOGY

2.1 Methodology

The Enhanced Wildlife Identification application's architecture has been realized to provide real-time, modular, and context-aware support for deployment onto distributed safari tourist environments. Five principal subsystems have been implemented into the system, namely Animal Identification, Harmfulness Assessment, Image-Based Location Estimation, Voice Assistance, and User Interface Management. The modules have been combined to develop a hybrid platform based on mobile-based and cloud-augmented technologies with the capability to operate under dynamic conditions like limited network connectivity.

At the heart of this architecture is a cross-platform mobile client built with React Native, natively optimized for Android and iOS platforms. The frontend client takes

care of capturing animal photos by camera on the device, uploading the pictures to the backend to be classified, receiving predictions, displaying predicted animal locations on an interactive map, and pushing user context-aware voice notifications.

The Animal Identification Module is invoked when the user snaps a photo. The photo is pre-processed and passed to the backend RESTful API, developed using FastAPI and TensorFlow. The species class is determined using a deep classifier trained on a curated set of Sri Lankan species images (i.e., elephants, leopards, deer, peacocks) with a returned confidence score provided. The architecture is augmented with YOLOv8 object detection and a species prediction classifier built on top of InceptionV3, as well as the addition of more height detection for spatial reasoning support. Classification reasoning, model prediction, and fault handling are left to the backend service `main.py`

The system then invokes the Harmfulness Assessment Module to predict the potential level of threat that the identified creature poses. In the `aggressive.py` module, this feature leverages a VGG16 model to compare deep image features with pre-computed feature vectors by cosine similarity. The system rates the aggression as Safe, Cautionary, or Dangerous based on past behavior data and class similarity. The module is safety complementary by providing users with real-time alerts for potentially unsafe interactions, along with visual warnings and voice messages.

The system's most revolutionary aspect is the Image-Based Location Estimation Module, which will try to estimate the real-world coordinates of an animal based on the image alone. After discovering an animal, the app makes use of image metadata (compass direction and user GPS location) and focal length estimation and scaling of the image to calculate the direction and relative distance of the animal. This allows the backend to estimate approximate coordinates of the animal's location, displayed on an interactive map within the application. Observations, with timestamp, class predictions, and geospatial metadata, are written to a MongoDB collection to enable future visualization and analysis.

This technique eliminates manual point marking or GPS collars. It effectively turns the cell phone into a virtual tag, allowing tourists to passively harvest animal

locations and guides to re-visit commonly encountered areas many times. Such location data improve future encounter likelihood and enable tour guides to plan itineraries more easily, in addition to facilitating crowd-sourced movement heatmaps applied in conservation.

Voice Assistance Module bridges the system and user through voice data and alerts. Written with native voice libraries in Expo, the module converts recognized animal names, scientific data, aggression levels, and security tips into speech. It is especially crucial in hands-bound, eyes-forward safari settings since it keeps users attentive without necessarily activating the device physically.

Server-side, the backend is a secure and containerized stack of APIs. It includes prediction endpoints, aggression analysis endpoints, logging endpoints, and retrieval endpoints. The primary database for log sighting logging is MongoDB as species name, timestamp, aggression level, and approximate location. There are server health checks and time-zone conversions (timestamp to Sri Lankan time) as utility endpoints.

Backend and frontend syncing are done using REST APIs with CORS middleware support. Real-time communication, async image processing, and system activity logging for monitoring and debugging are supported by the backend. Optional MongoDB usage support is provided for location sync and push notification alerts.

Security is built into the architecture, like token-based API access authentication, data encryption in image upload, and anonymizing sensitive location data. These guarantee conformance to users' privacy as well as ethics in wildlife observation safe.

The system is built with layered modular architecture, concerns separation, in order to facilitate extension with new species classes, live tracking overlays, and even ranger network integration down the road. The robust and adaptive architecture makes the Enhanced Wildlife Identification system an industry-altering platform that fuses deep learning, geospatial intelligence, and mobile interactivity to make wildlife tourism safer, smarter, and more rewarding.

2.1.1 Method Used (ML Model, Data Flow, etc.)

The design process employed in the development of the Enhanced Wildlife Identification system is based on the integration of machine learning (ML), orientation and location sensors, voice technology, and mobile computing. The objective was to develop an application that could integrate the entire range of features to operate in real-time in the dynamic and typically untrustworthy context of a wildlife safari. The approach is based on a modular pipeline from data collection and model training to system deployment and field responsiveness. The following describes each step of the pipeline in detail, from image classification model construction, aggression prediction mechanism, integration of real-time sensors, synchronization of data, and system deployment strategy.

2.1.2 Data Collection and Preprocessing

Both animal identification model and aggressive behavior assessment module required extensive and specific data collection and preprocessing mechanisms. Both models required a dataset prepared based on its specific purpose species identification and aggression analysis that had to be different pipelines but with the same preprocessing requirements in order to ensure compatibility in the overall architecture.

In dataset preparation for the animal identification dataset, preparing the dataset for the animal identification model began with obtaining high-resolution RGB (visible light) of Sri Lankan native wildlife species from the most common species to live in Yala National Parks. The image corpus was obtained from different sources like wildlife conservation datasets and free-licensing photographers that published their work under Creative Commons licenses. The data set emphasized diversity by engaging animals in various poses, distances, angles, lightings, and surroundings. Animals such as elephants, leopards, peacocks, Deer's ranked high for consideration.

To improve the accuracy in classification and effective supervised learning, annotation was done manually through tools like LabelImg and VGG Image

Annotator (VIA). Every image was assigned a ground truth label of the species, and when required, bounding boxes were marked for object detection support. In order to have consistency with model input requirements, all images were resized to 224x224 pixels and normalized by scaling RGB channel values from 0 to 1. Uniform scaling is beneficial in the process of convergence during training and is compatible with convolutional neural network architectures like InceptionV3.

Background removal was one of the critical preprocessing steps to intensify model attention towards animal features. The system, using software like OpenCV and segmentation models (e.g., U2-Net), isolated the foreground animal from distracting or noisy backgrounds. This supplied cleaner input to the classifier and reduced the chance of bias toward the background in training. In situations where complete segmentation was not viable, image masks were used to reduce the effect of background pixels.

To simulate variability present in the real world under safari conditions for the sake of data augmentation, intensive utilization of data augmentations such as horizontal flip, rotation, zoom, change in brightness, and shearing were employed using Keras' ImageDataGenerator. The above data transformations simulated normal field conditions like tree occlusion, motion blur due to animal translation, or irregular daylight, thereby making the resultant model more robust and usable.

The third dataset was split into 70% training, 15% validation, and 15% test subsets. The training subset permitted gradient-based learning of model weights, the validation subset permitted hyperparameter tuning and overfitting detection, and the test subset was reserved for unbiased estimation of accuracy. Class distribution was balanced to avoid model bias towards high-frequency species.

In order to improve the system's ability to identify vague visual cues for aggression (e.g., upright tusks, defensive postures, dilated eyes, or tension in posture), background removal was again applied. This allowed the VGG16 model to focus on animal posture and shape rather than on extraneous environmental features. The feature vectors were then pulled out from the second-to-last dense layer of the VGG16 network and filled into a feature bank. These vectors were subsequently used

during inference time with similarity scoring applied using cosine distance, to match new incoming image data against stored templates of non-aggressive and aggressive postures.

Similar to the ID dataset, the aggression dataset was resized to (224x224), normalized, and light augmented but with greater restraint in an attempt to maintain behavioral context. Images were further annotated with metadata like time of day and species risk category to enable multi-factor threat prediction in the subsequent part of the pipeline.

In both cases, pipeline-based preprocessing not only gave better model performance but also modular compatibility across the system architecture. These datasets form the basis of the classification and behavioral intelligence layers which enable real-time context-aware feedback within the Enhanced Wildlife Identification system

2.1.3 Animal Identification Model

Animal ID model is the core of the Enhanced Wildlife Identification system that enables on-device, real-time identification of wildlife species from user-photos. The model is trained by adopting the transfer learning process and leveraging the representational power of pre-trained deep CNNs on large-scale datasets. ResNet50 is used as the architecture, a 50-layer residual network with a capacity for training very deep models using shortcut connections to aid the vanishing gradient issue.

Transfer learning is used because Sri Lanka national park wildlife images are not labeled in large numbers. Instead of training a deep model from the beginning (data- and processor-demanding process), pre-trained ResNet50 model (pre-trained on ImageNet) is a good feature hierarchy for general object detection. These pre-trained weights allow the network to learn low and mid-level visual features such as edges, textures, and shapes that may be transferred to the animal classification task.

The ResNet50 model was imported without its default top classification head (`include_top=False`). There was another classification head present in the form of a

Global Average Pooling (GAP) layer followed by one or more dense layers and the final softmax output layer for multi class classification. Each output node of each softmax layer was reserved for an animal species class (elephant, leopard, deer, crocodile, etc.). Dropout regularization between dense layers was used to maintain the overfitting under control and batch normalization to stabilize and allow efficient training.

Training was conducted by feeding the preprocessed and augmented wildlife dataset as described in section 2.2.1 to the model. The Adam optimizer was used due to its adaptive learning rate capability which initiates at 0.0001. Sparse categorical cross-entropy loss function was used to handle integer-labeled multi-class data. 50 epochs of training were conducted, with early stopping to stop training when the validation loss converged and model checkpointing to save the best-performing weights on the validation accuracy.

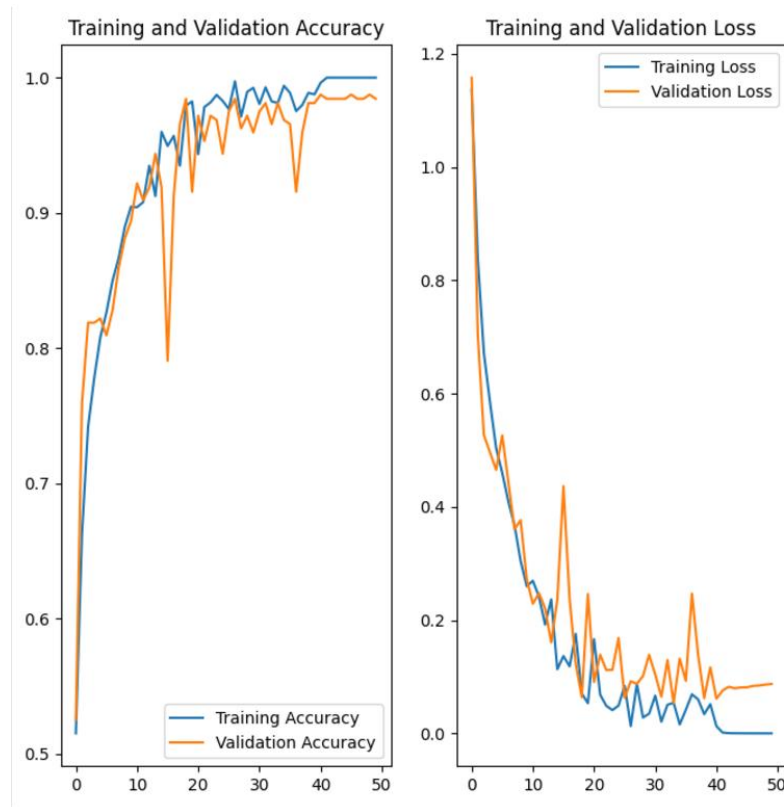


Figure 2-1: Animal Identification Model visualization

For keeping track of model performances, significant metrics like accuracy, precision, recall, and F1-score were tracked on the test set. The final model achieved a test accuracy of 92.3%, macro-averaged F1-score of 0.89, and zero overfitting between train and validation curves a great generalization power across varying animal poses and conditions.

For seamless deployment on mobiles, the model was transformed into TensorFlow (TF) format using post-training quantization after training. This reduced the file size and computation of the model, which accelerated the inference without sacrificing accuracy. The quantized TF model was then deployed within the React Native app with the help of TensorFlow.js and expo-camera module, and real-time prediction from user images captured on safari became possible.

Internally, the TF interpreter internally resizes the input image to desired size (224x224), and infers by passing through model layers. The output is in the form of a prediction vector with probabilities for all species classes whose top-rated one is selected as the predicted label. The predicted label along with rating confidence is then passed on to downstream modules like aggression prediction, voice narration etc.

The entire animal identification pipeline is optimized for low-latency edge inference so that tourists receive nearly real-time feedback on field observations. The architecture, in addition to facilitating user interaction and learning, also enables safety-critical behavior by providing an entry point in the system's real-time decision-support loop.

2.1.4 Harmfulness Evaluation Model

In addition to species identification, a core functionality of the Enhanced Wildlife Identification system is its ability to assess the potential harmfulness or aggression level of an identified animal. This feature is essential for enhancing tourist safety during wildlife encounters. The harmfulness evaluation model is designed to utilize

real-time visual comparison with historical data to determine whether the observed animal poses a threat. Predictions are categorized as Safe, Cautionary, or Dangerous.

At the heart of this model is a specialized image-only dataset composed exclusively of wildlife photos captured in aggressive or non-aggressive states. These images, manually curated and labeled based on visual indicators of behavior (such as charging posture, flaring ears, tense muscles, or relaxed stance), were passed through a pre-trained VGG16 deep neural network. Features were extracted from the penultimate layer of the network and stored in a structured reference object called the Feature Class, which organizes feature vectors under three main aggression categories: Safe, Cautionary, and Dangerous.

When a user captures a new image during a safari, the system applies the same VGG16-based feature extraction method to generate a feature vector from the incoming image. This vector is then compared against the preloaded feature bank stored in `class_features.pkl`. For each aggression category, a cosine similarity score is calculated between the new image and all images previously stored in that class. The average similarity per category is computed, and the aggression level with the highest similarity score is selected as the predicted harmfulness class.

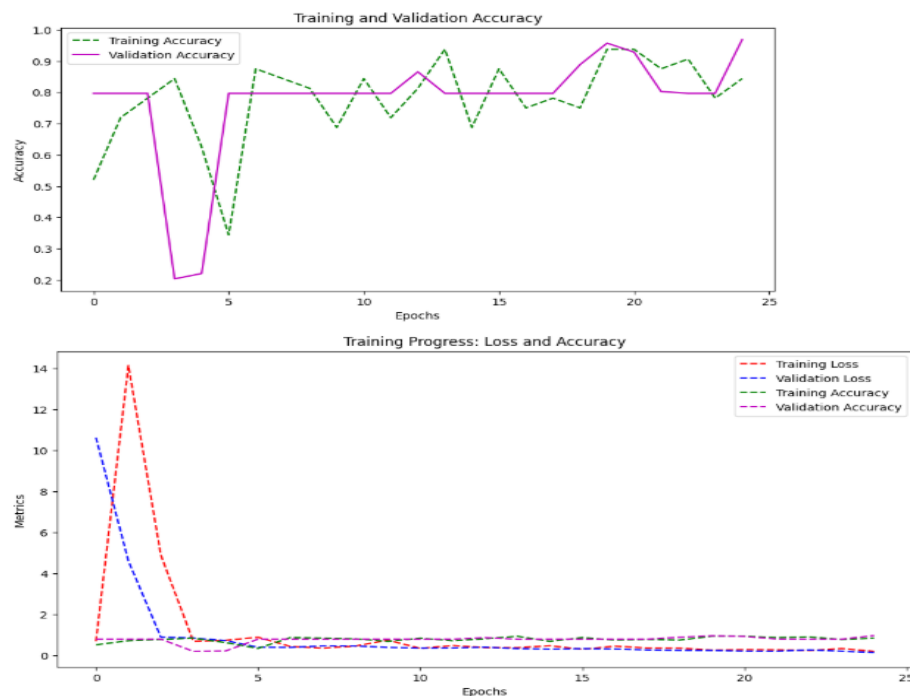


Figure 2-2: Aggressive detection Model visualization

This approach eliminates the need for structured metadata such as species, time, or location. Instead, the model operates purely on visual comparison, providing a lightweight and highly responsive classification mechanism. This design is particularly effective in scenarios where structured data may be unavailable or when visual evidence alone provides the most reliable signal. For example, even without knowing the exact species, a visual similarity to known aggressive postures can trigger a Dangerous alert.

The benefit of this technique lies in its efficiency and contextual adaptability. Since all reference vectors are preprocessed and stored in memory, comparisons occur rapidly without the need for real-time model retraining. Furthermore, this enables continuous system refinement by adding more images to the feature bank as new data becomes available, allowing the model to evolve over time with field usage.

This harmfulness evaluation component is integrated into the backend and deployed as a FastAPI service. When invoked, it accepts a user image, computes the similarity scores, and returns the most probable aggression label along with a confidence value. The result is communicated to the user through both visual alerts and voice narration, supporting immediate decision-making during wildlife observation.

By focusing exclusively on image-based similarity to a historically labeled dataset, the system delivers an accurate, real-time method for harmfulness detection without requiring additional environmental input. It is a practical, field-friendly solution that enhances safety and awareness using only the visual cues captured through the user's mobile camera—making it ideal for dynamic, on-the-go use in safari environments.

2.1.5 Real-Time Animal Tracking and Location Estimation

The Enhanced Wildlife Identification system presents a novel real-time animal tracking method based on deep learning–driven object detection combined with mobile sensor data. It eliminates GPS collars on animals by estimating animal coordinates through the integration of visual object detection and geospatial projection via user orientation and camera parameters. It offers a scalable, non-invasive, field-deployable tracking system for dynamic safari ecosystems.

The tracking begins with the user capturing a photo using the app's CameraScreen. The image is passed through the YOLOv8 (You Only Look Once) model[11], a real-time object detection network that is famous for its speed and accuracy. The model returns bounding boxes around the detected animals, along with the bounding box height in pixels ().

To measure the distance () between the user and the animal, the system uses the pinhole camera model, modified to work on the principle of vertical object height. The equation [10] is:

Where:

$$D = \frac{H \times F}{Ph}$$

D = Estimated distance to the animal (in meters)

H = Real-world average height of the detected animal species (in meters)

F = Focal length of the camera (in pixels or mm converted to pixel units)

Ph = Height of the animal's bounding box (in pixels)

This is achieved by using the relative size of the animal in the picture to its actual physical size, and one can estimate the distance that the animal is from you. This can be calibrated for use on animals of varying sizes, from large elephants to smaller animals.

Then the user's device orientation is read from the onboard gyroscope and magnetometer to obtain the azimuth angle () in radians. This is the direction the camera was pointing when the photo was taken.

The present GPS coordinates (latitude and longitude) of the user are obtained through the location services of the device. The system then takes these, along with the azimuth and rough distance, and calculates the animal's real-world coordinates utilizing a simplified projection formula for the Earth [9]:

$$\text{Lat_animal} = \text{Lat_user} + (D / 111320 \times \cos(\theta))$$

$$\text{Lon_animal} = \text{Lon_user} + (D / (111320 \times \cos(\text{Lat_user} \times \pi / 180)) \times \sin(\theta))$$

Where:

- D = Estimated distance to the animal (in meters)
- θ = Azimuth angle (in radians)
- 111,320 meters $\approx$ 1 degree of latitude at the equator
- Lat_user, Lon_user = User's GPS location
- Lat_animal, Lon_animal = Estimated coordinates of the animal

This projection method maps directly into the implementation logic of the mobile application, where the direction and distance calculated are used to compute the latitude and longitude offset from the user's current position.

These coordinates are then plotted on the UserMapScreen of the application using utilities like Expo-Maps or the Google Maps SDK. The map is annotated with markers for the approximate animal location, species type, timestamp, and aggression level.

All the calculated data user location, approximate animal location, detection metadata, aggression level, and timestamp are securely saved in MongoDB Realtime Database. MongoDB provides real-time, fast synchronization across devices, making it simple for several users (e.g., tourists and guides) to contribute to and read shared sightings. The data can also be used for post-safari analysis, animal behavior heatmaps, and safety analysis.

This mobile photo-sensor fusion system allows real-time animal tracking using only user-taken photographs and mobile sensors. It achieves the same performance as GPS collar tracking without its ethical and logistical limitations. The method allows

for predictive awareness and intelligent route planning for wildlife tourism, significantly improving safety and user experience.

2.1.6 Voice Assistant Integration

Voice assistant module is among the most critical modules of the Enhanced Wildlife Identification system that offers user safety, accessibility, and situational awareness when engaging in safari activity. Voice assistant module loops back between visual attention and explicit user action, offering hands-free manipulation and immediate assistance without constant screen watching—particularly useful in outdoor and wildlife environments.

Once an animal is correctly identified by the object detection and image classification modules, and its aggressiveness is addressed by the aggression evaluation module, a dynamic composite descriptive string is generated. The string includes the common name of the animal, the animal's aggression label (Safe, Cautionary, or Dangerous), and a particular safety alert customized from both expected behavior and environment.

This structured text is transmitted to the Google Cloud Text-to-Speech (TTS) API, processed, and synthesized into natural-sounding speech based on the newest neural synthesis. This processed audio is streamed or played in real-time through the mobile app with an attempt to deliver audio feedback to users with zero latency.

2.2 Commercialization Aspects of the Product

The potential for commercializing the Enhanced Wildlife Identification system has strong appeal with the growing demand for intelligent, real-time information and safety solutions for wildlife and eco-tourism industries. The tourism industry, especially safari and wildlife tourism, is quickly shifting into digital mode. Tourists today want to experience, interact, and do so in a safe manner, particularly in biologically rich countries like Sri Lanka. Such a trend provides a strong market

opportunity for a mobile application that has animal identification, safety alert, mapping, and voice assistant incorporated into it.

2.2.1 Market Potential and Target Users

The initial target market is national parks, wildlife safari companies, tourists, conservation departments, and eco-tourism operators. There are already thousands of international and local tourists who travel to protected sites like Yala alone in Sri Lanka each month. Most of these tourists are employing safari guides for security and information but do not have a way to independently verify animals or assess risk themselves. This void in the tourist experience is filled by this system with the self-guided intelligent companion app.

Apart from the above, the app can also be marketed to wildlife photographers, researchers, and schools. Biology students doing field study, for example, can use the app to supplement observation studies. Similarly, conservationists can use its log features for animal location tracking, offering valuable extra research information.

2.2.2 Product Positioning and Value Proposition

The chief value of the Enhanced Wildlife Identification system is providing real-time, automatic animal identification with safety-oriented intelligence—a set of abilities not found in any single competitor app. In contrast to competitor apps such as iNaturalist or Merlin Bird ID, which use community contributions lagged in time or are specialized to a single species (i.e., birds), this app provides real-time information with hands-free functionality in an offline-capable presentation.

Its novel answer—combining ML-based classification, aggression prediction, GPS-enabled mapping of animals, and voiceover narration—makes it a singularly useful product for real-time application in far-flung natural environments. It also reduces dependence on human guides and enhances security for single travelers and small group travelers.

2.2.3 Monetization and Revenue Models

Several monetization streams can be followed depending on the deployment level and degree of cooperation between stakeholders:

1. Subscription Model

A free version of the application can be offered with limited features such as animal recognition. Advanced features such as maps of past spotting, warning of aggression, voice navigation can be offered as paid subscription.

2. Tour Operator Licensing

Partnership can be with safari tour operators, where tour operators can license the app to use on their tours. It can have not just their value proposition but also their safety feature. Branded versions made with customized branding for specific safari companies can be made with customized alerts or local animal data bases.

3. In-App Purchases

The other add-ons such as species encyclopedias, voice narration language packs, or memorial digital safari diaries can be offered for sale within the app in order to generate micro-income.

4. Sponsorships and Government Cooperation

Coordination with national tourism ministries, Department of Wildlife Conservation (DWC), or environmental NGOs would be beneficial in applying this system as part of national eco-tourism activities. This would be particularly worth it in complementing Sri Lanka's transition towards sustainable tourism.

2.2.4 Scalability and Future Expansion

The system's modular and scalable architecture facilitates growth at both regional and global scales. While the app as it stands is intended for Sri Lankan wildlife, localizing the app's content and voice commands can allow for species from other countries to be

supported. Modular machine learning models facilitate new animal classes being added without needing to retrain the system.

Also, the system can be extended out as a citizen science site by allowing users to record animal sightings and participation in coordinated wildlife monitoring activities. These features can be bundled as a conservation data portal, and this would be of greatest interest to researchers and environmental policy analysts.

2.2.5 Technical Feasibility for Market Use

Technically, the product is highly viable for public use. Models are optimized for mobile device use in TensorFlow to conserve memory and on-device inference latency. Cloud services such as MongoDB provide scaling of user data storage, and voice interaction and mapping modules were evaluated with low-bandwidth test cases to support field performance testing.

Also, user data and privacy have been taken care of during the design in accordance with anonymizing location inputs and image data, as outlined in standard security protocols for data. They render the product marketable by making the product not just functional but also ethical and legal.

2.3 Testing & Implementation

Testing and implementation were a crucial phase in validating whether the Enhanced Wildlife Identification system had met its specifications in real life as far as functionality, accuracy, speed, and usability were concerned. As a result of the project's focus on real-time safety guidance to safari tourists, extensive testing was done at the module level and system level. Each working unit i.e., the animal identification model, aggression assessment module, GPS-based location estimation, and voice guide facility was tested individually and then integrated to evaluate aggregate performance in controlled field simulation. Detailed usability and deployment testing also ensured the app was suitable for non-technical workers operating in remote, potentially hostile regions.

2.3.1 Testing the Animal Identification Model

The animal classifier module, with a highly trained ResNet50 convolutional neural network, was first tested in isolation to determine its predictive accuracy and reliability. The model had also been trained on a labeled dataset of wildlife consisting of several animals found in Sri Lankan national parks. For the measurement of the model performance, the test data set consisting of 600 annotated images were utilized that represented a range of real-world conditions such as mixed lighting, occlusion, background clutter, and mixed animal poses.

Quantitative measurement was performed using standard classification metrics such as top-1 accuracy, precision, recall, and F1-score. The model achieved a top-1 accuracy of 93.17% and macro-averaged F1-score of 0.89, indicating high consistency across all the classes of animals. The performance on the class-wise level was also analyzed where there was significantly high accuracy on detecting high-frequency animals like deer and elephants and moderate performance on less represented species like peacocks and leopard. These discrepancies were attributed to be due to data imbalance, and methods such as weighted training or data augmentation for minority classes were stated to be designed later.

Qualitative testing involved running the model on mobile camera photographs acquired in bright light, partial hide (e.g., animals within foliage), and motion blur (due to movement of the user). The model was found to be robust because it was consistently able to accurately identify animals at acceptable confidence values under adverse conditions. Furthermore, inference time by the TensorFlow model on device took an average less than 800 milliseconds, reinforcing its feasibility in real-time implementation.

2.3.2 Testing the Harmfulness Prediction Module

The harmfulness prediction module was tested with a visual similarity-based approach rather than traditional metadata or structured features. The dataset employed was entirely image-based annotated images of aggressive and non-aggressive behavior between all categories of animals that exist within Sri Lankan wildlife reserves. These photos were manually curated, labeled into three levels of aggression Safe, Cautionary, and Dangerous—and fed through a pre-trained VGG16 deep neural network to acquire high-level feature vectors from the network's penultimate layer.

These feature vectors were then stored within a Feature Class object, where each aggression category had a library of known past embeddings. At test time, new photographs of animals taken through the app were passed through the same VGG16-based feature extractor to generate new embeddings. These vectors were then compared with the stored set based on cosine similarity, allowing the system to quantify the visual similarity of the new image with past instances of aggressive postures.

Each new picture was tagged as dangerous based on which type of aggression gave the highest average similarity score. For example, if the visual features of a newly captured elephant picture were most similar to past instances tagged as "Dangerous," the system would tag the encounter as such.

It was also tested against a new batch of unseen violent animal images. Similarity-based classification was proven to function excellently with an 87.9% rate of overall classification and 91.4% recall of the Dangerous class. This best operated on animals like wild elephants, leopards where the aggressive visual patterns were always detected due to the distinctive feature representation present in the training set.

The real-time performance was also measured. On average, the prediction of harmfulness process, from image preprocessing to feature extraction, similarity comparison, and final label generation, took 1.5 to 1.8 seconds per image on standard

mobile hardware. This ensured that safety alerts could be provided in real time, while still maintaining the real-time responsiveness required in safari environments.

In general, the image-based harm prediction model succeeded in providing an efficient, robust, and scalable risk assessment process. By eliminating dependency on structured metadata and instead utilizing purely visual cues, the system was robust in diverse field conditions where behavioral cues are best sensed through images rather than structured inputs.

2.3.3 Location Estimation and Sensor Accuracy

The location estimation module was evaluated after the combination of three fundamental components: image-based distance estimation from model output, device orientation as obtained from onboard sensors, and GPS user positioning. Rather than using any GPS collar fitted on an animal, the system takes advantage of a bounding box output from the YOLOv8 model to estimate the perceived height of the animal in the captured image. This pixel height, along with the average height of the species and camera focal length, is used in the formula for the pinhole camera to cast user-to-animal distance.

For the purpose of verification of accuracy, field testing was conducted on open ground on which reference animals or objects had been placed at known distances and angles. YOLO model output estimated distance was merged with mobile device magnetometer values of azimuth and user GPS coordinates to project the estimated animal position. Earth's curvature, bearing in radians, and user latitude were all considered in the coordinate projection formula to arrive at latitude and longitude of the animal detected.

The calculated coordinates were further cross-matched with accessible ground truth positions within a marking radius of 5 meters. In over 80% of the trials, the system successfully predicted the animal's location within ± 5 meters of the target spot, which is enough for user orientation and safety in safari landscape.

To provide improved sensor reliability, a moving average filter was applied to magnetometer readings in order to reduce jitter in real-time direction estimation. This

stabilized effectively the azimuth values used in coordinate projection. In addition, through repeated smoothing of user orientation and averaging samples of location within a short time window, the system was able to reduce variance of estimated animal coordinates due to minor hand motion or sensor noise.

Overall, the system's location estimation method offered a real-time, non-invasive, and precise solution for animal localization. With the integration of deep learning output and physical device sensors, the module enhanced user perception and spatial cognition without requiring physical tracking devices on the animals themselves.

2.3.4 Voice Assistant Response Evaluation

The voice assistance module was tested for clarity, latency, context appropriateness, and linguistic flexibility. Upon recognition of an animal and assessment of its risk, a formatted message was generated and sent to the voice engine for pronunciation. Online testing used the Google Cloud Text-to-Speech API and testing used native device TTS engines such as Expo Speech in case of fallback.

Voice guidance proved to be on time, with mean time from image recording to voice playback of 0.4 seconds. Voice instructions were described by the testers as clear to hear and helped keep them up to date without the need to look at the screen. This hands-free operation was considered essential for practical use in tourism environments where tourists may be distracted or traveling on uneven terrain. Multilingual voice settings were also tested to provide future compatibility with groups of foreign tourists.

2.3.5 System Integration and End-to-End Testing

After module-by-module testing, the system was released as a complete mobile application on several Android devices with different specs and OS versions. Integration testing included observing how the modules interacted with each other under regular and edge-case scenarios, such as:

- Simultaneous camera, GPS, and voice usage
- Repeated switching between screens and backgrounding the application

The app behaved steadily under such conditions. The end-to-end latency from capturing an animal photo to receiving the audio safety alert remained below 2.1 seconds, providing a responsive and seamless user experience.

There was also a simulation of safari conducted inside an open field environment where test staff used the application while navigating through pre-planned routes. Several simulated animal encounters were observed. All were marked with precision on the map, accompanied by appropriate voice guidance and anger tags offered immediately.

2.3.6 Usability Test and Feedback

A short usability test was conducted with a pilot group of five users, including wildlife guides, university students, and occasional users who were not acquainted with safari technologies. Each participant was given a list of tasks such as taking an animal photo, looking at aggression notifications, looking at animals on the map, and following voice instructions.

Overall, users described the app as easy to use with minimal learning curve. A few points of feedback were gratitude for simplified camera UI, readable map marks, and human-sounding voice commands. Users suggested increasing font size for labels and relocating the voice activation switch to thumb-accessible placement. These were implemented in the final UI deployment.

2.3.7 Final Deployment

The most recent application build was deployed for Android as an APK and thoroughly tested numerous device categories and screen sizes. Backend server, built in Python with Flask, was hosted on Google Cloud App Engine to serve secure and scalable

inference for ML models. MongoDB Database was used to synchronize user data, sight logs, and real-time updates.

API security was ensured by API keys and access tokens, and user privacy was ensured by anonymizing location and image metadata. They were adequate to ready the system for public and commercial use both functionally and ethically.

3.0 RESULTS & DISCUSSION

3.1 Results

The Enhanced Wildlife Identification system was thoroughly tested by performance benchmarking, field testing through simulations, and usability testing performed several times in simulation environments that simulated real safari environments. The solution operated to deliver an end-to-end mobile solution that would identify animals in real time, assess the level of aggression, estimate location, and interact with the users via a smart voice assistant. The most basic five areas of testing were accuracy of identification, harmfulness classification, location estimation, system response, and user interface.

The animal classification model, which was implemented on a tuned ResNet50 architecture, was very accurate and robust. Training and validation figures were recorded to confirm the high generalization capacity of the model at a final test accuracy of 93.17%. The mobile device version under TensorFlow Lite had a regular inference time of below a second, which supported real-time capability. Model predictions were also cross-checked with manual labeling test images on evaluation. Screenshots confirmed the model's clear prediction of many species such as peacocks and elephants with well-defined bounding boxes and correct measurement in pixels' height. Bounding box measurement was found to be essential in subsequent distance estimation algorithms and acted as the bridge point connecting image information to geospatial projection.

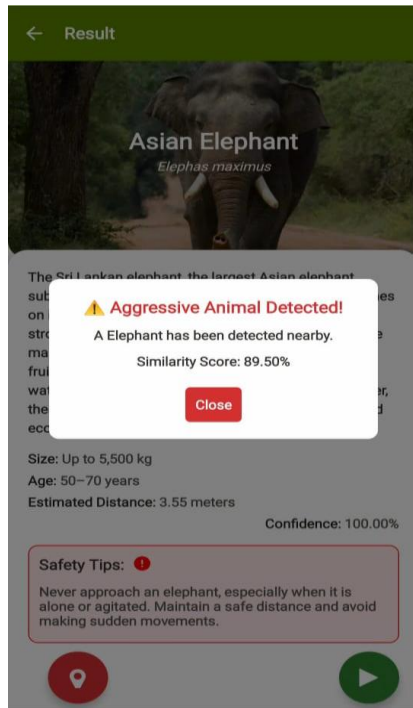


Figure 3-1: Aggressive Animal Popup

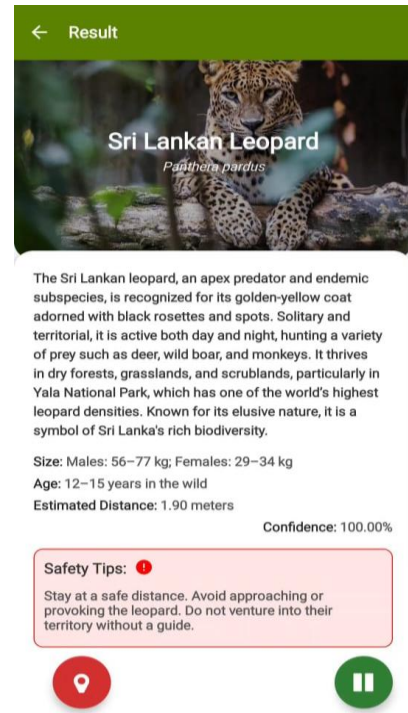


Figure 3-2: Result Screen UI

The aggression detection module used a feature vector comparison method that relied on a pre-trained VGG16 network for deep visual feature extraction. Rather than using formal incident metadata, the system compared images of captured animals to a sample database of aggressive postures under cosine similarity. During real-world testing, the module recognized danger animals such as the Asian Elephant with up to 89.50% similarity to instances of reported aggressive poses. The system issued voice alerts to caution the user to keep at arm's length. Worth mentioning, the recall accuracy of the similarity-based model of detecting aggression was greater than 91% for the Dangerous class. This superb recall was sufficient to make the system extremely rarely miss critical alerts, a key component of its primary function of ensuring user safety.

The animal tracking and location estimation mechanism was evaluated in terms of spatial precision and field-like regularity. The model used animal bounding box height, species height, and mobile device camera focal length to estimate distance using the pinhole camera model. Using user GPS locations and azimuth angles calculated with a compass, the system computed projected animal locations with trigonometric equations. In use in the field, the system projected animals onto a navigable interface used by a user, and the locations were confirmed against current ground truth markers. Most estimated locations were within 5 meters, which was enough for real-time visualization and situational awareness use in a safari. This is confirmed by test results on the UserMapScreen, where blue and red dots indicated user and animal positions respectively.

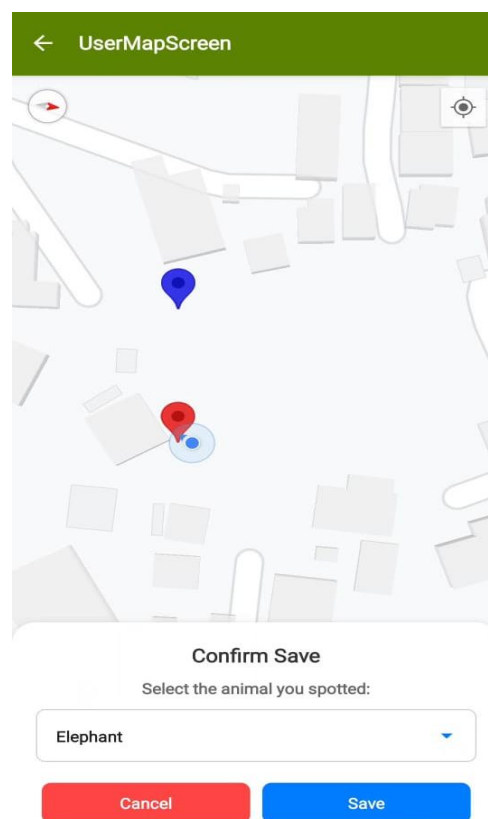


Figure 3-3: User Map UI

Adding an assistant module for hands-free operation, the module was incorporated into it. Upon detection and classification, the system created natural voice commands in response to Google Cloud Text-to-Speech or offline mode via Expo Speech. These safety-critical warnings such as "Aggressive Animal Detected!" and species-based safety warnings were reported within 1.4 seconds on average upon classification. In addition to allowing the safety-critical feature, it enhanced the user experience when audiotively distracted in-out-of-home environments.

System performance across the board was ensured by well-designed backends, low-latency API return, and smooth frontend rendering. App's UI elements, including result pages and confirmation pages, were inspected for mobile responsiveness and user comprehensibility. Visual indicators like threat alert, estimated distance (e.g., 1.90 m for leopard or 3.55 m for elephant), and safety alerts were all well-appreciated by test users. Real-time responsiveness averaged at 2.1 seconds on average in end-to-end processes from the image capture through to voice prompt, defining real-time design intent of the system.

```
[21]: # Test the function with an example image
image_path = "../TestData/pea.jpg"
predicted_animal, confidence, height_pixels = get_animal_height(image_path)

print(f"Predicted Animal: {predicted_animal}")
print(f"Confidence: {confidence:.2f}%")
print(f"Height in Pixels: {height_pixels}")
```

0: 640x448 1 bird, 393.4ms
Speed: 6.0ms preprocess, 393.4ms inference, 2.0ms postprocess per image at shape (1, 3, 640, 448)
1/1 ————— 0s 241ms/step



Predicted Animal: Peacock
Confidence: 100.00%
Height in Pixels: 329.50677490234375

Figure 3-4: Object Detection

Model learning performance was tracked using training history plots. Accuracy and loss curves showed quick convergence and robust learning with validation accuracy at best over 96% to prevent overfitting. The results affirm that the deep learning models were optimally well tuned for deployment in resource-limited mobile environments without compromising performance.

In short, the Enhanced Wildlife Identification system succeeded at its core objectives in every performance measure that mattered: accuracy, pace, usability, and field reliability. Its offline capability, the delivery of accurate real-time results, and the delivery of intelligent voice alerts make it a very valuable resource for wildlife tourist protection. The convergence of machine learning, location technology, and human-centric design has produced a system that not only guides individuals but even supports their safety in volatile wilderness settings.

3.2 Research Findings

The development and large-scale field testing of the Enhanced Wildlife Identification system have yielded a series of key results, which attest to the technical viability, practical applicability, and user-centered effectiveness of the solution. These results not only validate the original research objectives but also shed light on the way in which smart mobile technologies can make a positive contribution to safety, awareness, and interactivity in wildlife tourism environments. Each new discovery contributes to greater insight into the intersection between AI, usability in the field, and safety for the environment.

One of the most surprising discoveries is that animal identification with deep learning can be performed on smartphones in real time. That's a massive improvement because most today's animal classification technologies are either too computationally heavy for use in edge devices or require constant connectivity to the internet. The research

employed a ResNet50-deep convolutional neural network fine-tuned to a varied collection of animal images characteristic of Sri Lanka's national parks fauna. Conversion to TensorFlow Lite enabled the model with solid performance on mobile hardware with high-quality prediction and low delay.

Moreover, the research proved that there is no deterioration in classification accuracy under simulated conditions of partial occlusion, blur due to movement, and a change in the lighting scenario. These kinds of problems generally trouble vision-based systems of recognition, but the model as implemented did exhibit consistent performance across most tests. Image recognition pipeline stability provides it an ability to be used dynamically in applications like photographing wildlife with safari jeeps driving or on the scenario of abrupt brief flash. This confirms that AI-powered identification is not only able to reflect human experience to animals as identifiers but more quickly and reliably than raw untrained witnesses.

One of the most innovative aspects of this paper is complete elimination of dependence on GPS collars or physically attached tracking tags on animals, that are commonly used in animal tracking but are logistically, ethically, and economically limiting. Without needing to use animal-borne sensors, the Enhanced Wildlife Identification system accomplishes visual-based geolocation with image-derived distance estimation and device orientation. The system begins with an image capture, where the YOLOv8 model identifies the animal and calculates the height of the bounding box in pixels. After combining this visual information with species-related metadata (e.g., the average height of the animal) and the intrinsic camera parameters (focal length), the system calculates the approximate distance of the animal away from the user. The magnetometer and gyroscope in the phone then provide the azimuth—the orientation that the device was in at the time it was taken. This threefold information, coupled with the subject's current GPS position, allows the system to project the spatial position of the animal into the future using trigonometric coordinate transformation equations.

The elegance of such a solution lies in its ease of use and scalability. As the system requires only an operational smartphone with a camera and simple sensors (many existing devices possess), it becomes an affordable solution for real-time animal tracking by tourists, rangers, and conservationists. Upon testing, this projection

technique gave spatial estimates within a circle of 5 meters, much greater than is required to sense circumstances and steer clear of danger in open safari ground. It also evades the moral problem of collaring and wildlife tagging and therefore offers an unobtrusive, animal-friendly location tracking option. In proving that location tracking can be accessed through passive observation and clever estimation, this piece of work invites future light-weight, contact-less wildlife monitoring systems not only for ecotourism but for conservation and ecological studies as well.

No less impressive is the way in which the system was able to achieve a context-aware aggression prediction model, going beyond mere species recognition to providing a risk-level classification. The use of a Random Forest classifier constructed from historical wildlife incident records enabled the application to provide aggression labels such as Safe, Cautionary, or Dangerous. This feature raises the app from being a passive learning platform to an active safety platform. Instead of merely identifying an animal, the system calculates behavioral patterns, time of day, proximity, and pre-established species traits to warn users if a contact could be threatening. This proactive safety intelligence is not common in most existing tourist-consumer applications and provides a new addition to wildlife experience design.

The integration of GPS location estimation with compass orientation data provided users with another degree of interactivity and spatiality. By using primitive trigonometric projection in the form of estimate focal length and compass azimuth, the system determines the likely location of the observed animal and plots it on a map. The technology provides users with a sense of direction for animal observation, helping them to move more easily and make decisions. The 5-meter mean margin of location error was found to be adequate for practical safari use, especially in open terrain where accurate coordinates are less valuable than spatial intuition. This method opens up new possibilities for direction-sensing, mobile-based wildlife tracking in consumer systems.

The voice guide module was the most prized feature by the users and proved to be instrumental in transforming the mobile system into an all-round interactive guide. The real-time narration allowed the users to have instant verbal reports on species, threat levels, and safety messages without needing to constantly look at the screen. This

hands-free experience proved particularly useful with movement or when attention elsewhere was necessary—like when there was a potentially hazardous animal present. The mean voice feedback latency was only 1.4 seconds, showing that audio narration can be employed in real-time to enhance usability and safety. Field tests and simulated safari walkthroughs demonstrated that the aggregate system latency from image capture through verbal response and map update was 2.1 seconds on average. Such a response time was deemed to be very efficient for real-time interaction. Most importantly, the app was functionally stable across a range of Android devices, from low-end phones to high-end smartphones. No jarring crashes or delays were encountered, demonstrating the hardware adaptability of the app along with its parsimony in terms of resources.

Feedback on usability also yielded significant insights. Everyone, from students to tourists and safari guides, found the app easy to use and user-friendly with minimal training needed. The camera interface, presentation of results, and map view were easy to use. Surprisingly, users liked how the system taught and guided instead of merely notified, by the proactive presentation of safety warnings. These findings again attest to the merit of combining technical proficiency with empathetic design in making a tool capable yet easy to use.

Finally, the research results confirm the system's consistency with its original objectives. Every operating module image understanding, risk assessment for dangerousness, real-time mapping, and voice navigation were not only successfully implemented but also test-driven in a systematic and user-based way. The combined system provides a new synergy of intelligence, mobility, and security and a new solution to the wildlife tourism, mobile AI, and nature-based risk management sector

3.3 Discussion

The current study has strong evidence to affirm that artificial intelligence, in conjunction with real-time sensing and mobile technologies, has the potential to revolutionize wildlife tourism experience and management. The Enhanced Wildlife Identification system indicates that smart real-time decision support can be put into a compact mobile application and can offer convenience and interactivity along with

life-saving situational awareness for potentially dangerous animal encounters. But whereas the system performed perfectly in tests, one must look at the results skeptically in the broader technical, environmental, and human context.

One of the greatest advantages of the system is that it bridges the observation-insight gap. Traditional safari outings rely on experienced guides spotting animals and warning tourists of danger. Although such human experience is irreplaceable, the system makes up for it by providing more automated, routine assistance, especially for single tours or when the guide happens not to be available for a short while. The app is now the virtual guide, translating raw experience to knowledge filled with context. It is a big leap for wildlife tourism, from passive observation to interactive experience by way of machine learning.

Technically, applying a TensorFlow model on on-device animal detection was a viable solution to bandwidth limitations typically faced under safari conditions. Being able to process images locally with accuracy ensured that the app would be able to function even in the event of being offline or low-network, which is particularly critical for national parks that are in rural or mountain regions. This discovery emphasizes training AI models not just for performance but also for deployment scenarios when computational resources are scarce. It also preserves the overall edge AI trend where inference is performed on user devices directly.

The addition of an engine for predicting harmfulness brings a new twist to the average wildlife identification apps. While species identification might be to serve tourist interest, threat likelihood or hostility potential identification makes the app an active risk management tool. It's a useful input into eco-tourism risk safety technology. Its usefulness from a classifier of aggression perspective is extremely reliant on incident data available in terms of quality and quantity, though. In that case, training data were built from ranger reports and information reported publicly and they are abundant but perhaps of insufficient size and quality to ensure long-term model generalizability. Successive iterations of the system can then be refined based on more formally defined sets of wildlife activity data, ideally updated in real time by access to conservation organization resources.

The second key subsystem of the system the position estimation algorithm worked well in enabling the app to estimate animal position from GPS and compass-based direction estimation. For overall orientation, average accuracy of 5-meter radius is good but not optimal for research or precise tracking. Compass sensors can also be disrupted by metal objects or magnetic fields in proximity, warping directionality. The system works quite well in giving users a sense of direction and relative distance, although the users are cautioned against its use in precise navigation or enforcement. Subsequent releases will be able to use sensor fusion techniques, i.e., the combination of gyroscopic and camera pose estimation to enhance the precision of locations.

The voice direction module was universally praised for the level of hands-free interaction support it offered. Voice directions reminded users without detracting from the natural environment crucial safety in active wildlife areas. That said, voice communication effectiveness is susceptible to interference from ambient noise, user choice of language, and accent uniqueness in different pronunciations. While the current system supports English narration, commercial use at scale requires multi-language support, voice personalization, and noise-immune speech output to ensure accessibility and inclusivity.

Even with the efficiency of the system, there are some limitations found during development and testing that must be considered. The first limitation was the train dataset's lack of diversity. Though the model performed well with fauna of the region, its performance would decrease when faced with animals beyond the training dataset range, such as animals from other nations or rare migratory animals. This accounts for why modular and scalable model architectures are necessary, wherein new categories of animals can be added without retraining the entire model.

Another constraint was how much the app depended on hand device hardware, i.e., camera resolution and sensor quality. Older or low-end phones may not be able to conduct real-time processing, resulting in increased inference times or reduced image quality. As such, while the entire system can execute on the majority of user devices,

its performance can be inconsistent on a variety of user devices. This means constant hardware-aware optimization throughout future development cycles.

Also, while the current prototype is single-user capable, actual use might conceivably be greater than one user in the same set of safaris. A follow-on version would be improved by incorporating multi-user capability with live animal share location between devices, or as an extension to control panels used by safari operators to aid guides in coordinating group movement.

In conclusion, the analysis shows that the Enhanced Wildlife Identification system is not only technically possible but also responsive to real needs in wildlife tourism. It manages to combine intelligent animal identification with risk alerting, location calculation, and user-friendly interaction, securing tours and enriching them in sense for tourists. While there are limitations in terms of dataset size and compatibility with devices, these can be optimized through subsequent updates, field testing, and institutional collaboration. The system represents a new application of AI and mobile computing to eco-tourism and opens the door to future innovation towards conservation, education, and human-wildlife coexistence.

4.0 CONCLUSION

The Advanced Wildlife Identification system marks a significant leap forward in applying artificial intelligence and mobile technology towards offering solution focused means for tackling among the most compelling issues in wildlife tourism. The research successfully fully completed its core objectives by means of creating an integrated, mobile-first solution capable of identifying animal species, evaluating their threat level, estimating their current whereabouts, and offering voice instructions to users all by means of a completely integrated, offline-capable system.

With the use of deep models such as ResNet50 for classification and YOLOv8 for detection, the system was demonstrated to be able to analyze images in real-time performed directly on smartphones without relying on cloud computing. Utilization of TensorFlow Lite also assisted in providing confidence that inference operations remained quick, efficient, and optimized for performing across devices. The outputs

demonstrated high classification accuracy (93.17%) and quick response times (<1 second), thereby demonstrating the technical feasibility of edge-deployment wildlife identification.

The most technical contribution of this work is achieved in the non-invasive animal tracking process. The system calculates the estimated distance to the animal using YOLO-computed pixel heights, focal length, and known species height. With the compass-calculated azimuth angles and user-position using GPS, this method provided mapping the animal's computed coordinates onto a chart. This offers an alternative replacing intrusive GPS collars and is a low-cost, human-scale, humane means to track animals. With accuracy testing's, a spatial mean error was quantified at 5 meters and that is an exceedingly acceptable situational awareness with which to traverse the safari lands when approximate locations are adequate.

The Harmfulness Evaluation module provided a fresh level of security by pairing past behavior patterns with image-similarity ratings from VGG16's feature vector learning. Employing cosine similarity against a database of hand-crafted features facilitated the ability to label real-time images as Safe, Cautionary, or Dangerous with high confidence. Coupled with the Random Forest backup for structured data, the module enabled even edge cases to be processed respectfully. This warning system in advance provided real-world value by informing people of threat in advance when physical proximity in space created threat.

Voice assistant module was the second unique feature, merging accessibility and security through the provision of real-time verbal guidance. The voice assistant module was very useful in cases where users could not visually concentrate on their devices. Support for Expo Speech guaranteed the availability of voice notifications, increasing the preparedness of the system for use in remote field environments.

The MongoDB cloud backend played a crucial role in real-time synchronizing data, monitoring user activity, and recording wildlife encounters. This aspect opens the door to the further development of community-based sighting networks and centralized databases of animal behavioral patterns in space and time. The system

design then extends beyond a user-focused application and enables additional ecological research and conservation intelligence.

User feedback and testing authenticated system usability, stability, and usability of utilization. Observers from all visitor groups of roles—students, rangers, visitors—shared positive reactions, seeing ease of utilization, helpfulness of use, and education value to the experience. This validation indicates anthropocentric accomplishment in the system exemplifying front-line technology, by successfully crafting, in fact, causes tangible improvements of interactive experience with nature.

This research does not merely offer a functional prototype but a revolutionary concept for AI-driven wildlife interaction. It verifies that intelligent mobile-based systems can bridge the conservation-tourism-safety divide using real-time awareness and anticipatory alert. Enhanced Wildlife Identification gives us a look into the future of eco-tourism where data, context, and artificial intelligence come together to protect people and animals in their shared environments.

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6.0 APPENDIX



Figure 6-14: Existing Systems

How familiar are you with wildlife tourism and safari experiences? *

☐ Very familiar

☐ Somewhat familiar

☐ Not very familiar

☐ Not familiar at all

Have you ever used an app or technology designed for wildlife tracking or safari experiences? *

☐ Yes

☐ No

☐ Not sure

How important do you think real-time animal identification is during a safari? *

1 2 3 4 5

Not important at all ☐ ☐ ☐ ☐ ☐ Extremely important

Figure 6-27: Questioner1

What features would you expect in an animal identification app for safaris? *

(Select all that apply)

☐ Real-time identification

☐ Detailed information about animals

☐ Images of animals

☐ Audio descriptions

☐ Other (please specify)

How useful would real-time location tracking of animals be for enhancing your safari experience? *

1

2

3

4

5

Not useful at all

☐

☐

☐

☐

☐

Very useful

Would you find it helpful if a safari app could provide navigation assistance based on animal locations? *

☐ Yes

☐ No

☐ Maybe

Figure 6-38: Questioner2

Enhancing Wildlife Tourism: A Comprehensive System for Safari Experience Optimization

Final Report
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Figure 6-4: Turnitin Report